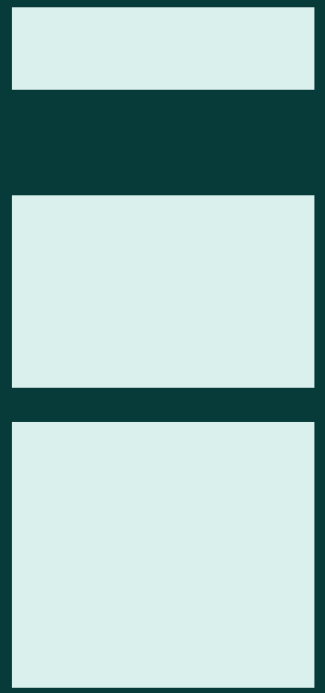
SWYNEX – E-commerce Sales Analysis

From Messy Data to Business Insights

END-TO-END DATA ANALYTICS PROJECT

Python · Pandas · Matplotlib · Power BI
Presented by Gautam · Synthetic dataset for practice purposes



CLEAN DATA
SMART ANALYSIS
BETTER DECISIONS

Made with GAMMA

From raw records to business insight

The project turns unreliable e-commerce data into a decision-ready view of sales, operations, and customer experience.

01	02	03	04	05	06
Raw Data	**Data Cleaning**	**Exploratory Data Analysis**	**Visualization**	**Power BI Dashboard**	**Business Insights**
Start with source records.	Remove noise and errors.	Find patterns worth testing.	Make evidence readable.	Turn analysis into a reusable tool.	Convert findings into action.

01

Clean and prepare raw data

02

Identify trends, concentration, and revenue leakage

03

Build an interactive dashboard and recommendations

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03 / DATA PREPARATION

Cleaning removed the noise

RAW RECORDS

281

→

cleaning

CLEANED RECORDS

252

SCOPE

19 columns

Jan 2025 – Aug 2026

ENTITY COVERAGE

15 / 15 / 6 / 8 / 7

customers · products · categories · cities · states

PYTHON + PANDAS

Handled missing values, duplicates, invalid quantities/prices/ratings/discounts/dates, inconsistent spellings, data types, and recalculated total amount.

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EDA revealed the patterns worth acting on

The analysis moved from description to diagnosis: status, timing, and revenue performance became visible.

01 **Orders by status:** cancelled, delivered, shipped, returned, pending

02 **Monthly revenue:** uneven performance with a sharp July 2026 peak

03 **Orders by weekday:** Sunday is the strongest ordering day

Also tested category, product, city, payment, rating, and IQR outliers.



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Electronics and one hero product drive the economics

TOTAL ORDERS
252

TOTAL ORDER VALUE
₹34.1 lakh

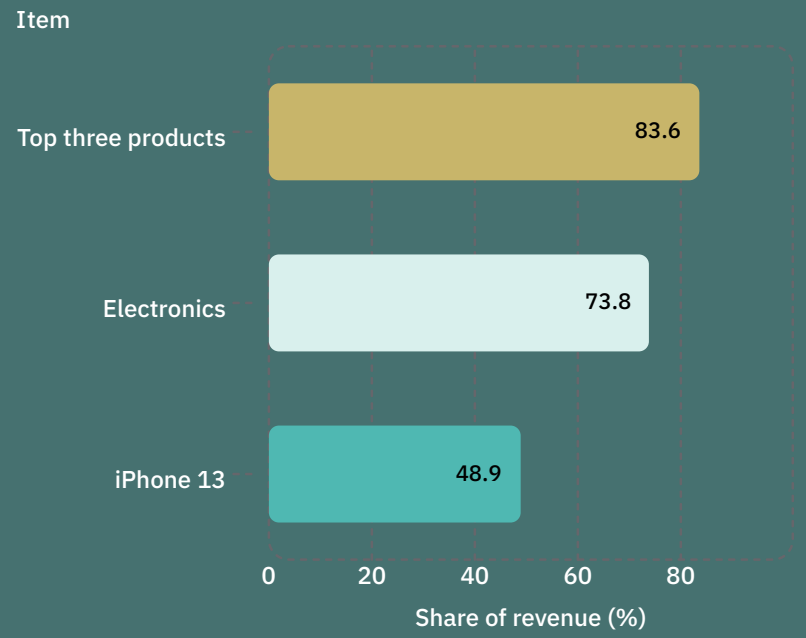
ACTUAL REVENUE
₹18.8 lakh

Excluding cancelled and returned orders

AVERAGE ORDER VALUE
₹12,768

AVERAGE RATING
3.07 / 5

DELIVERY TIME
5.4 days



Electronics contributes 73.8% of revenue but 22% of orders.

iPhone 13 contributes 48.9% of revenue; the top three products contribute 83.6%.

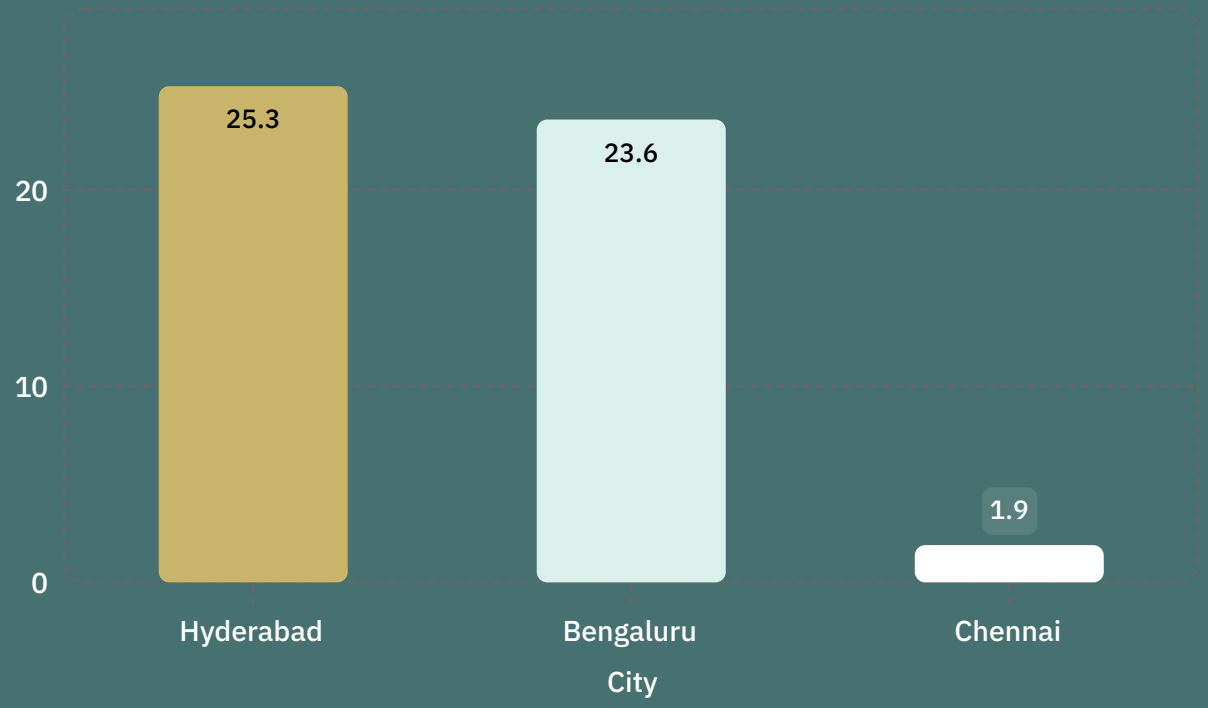
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06 / CONCENTRATION

Revenue is concentrated in two cities

Hyderabad: 25.3% of revenue · Bengaluru: 23.6%

Sunday: highest order count



TOGETHER

48.9%

of revenue

CHENNAI

1.9%

of revenue

OPERATING PATTERN

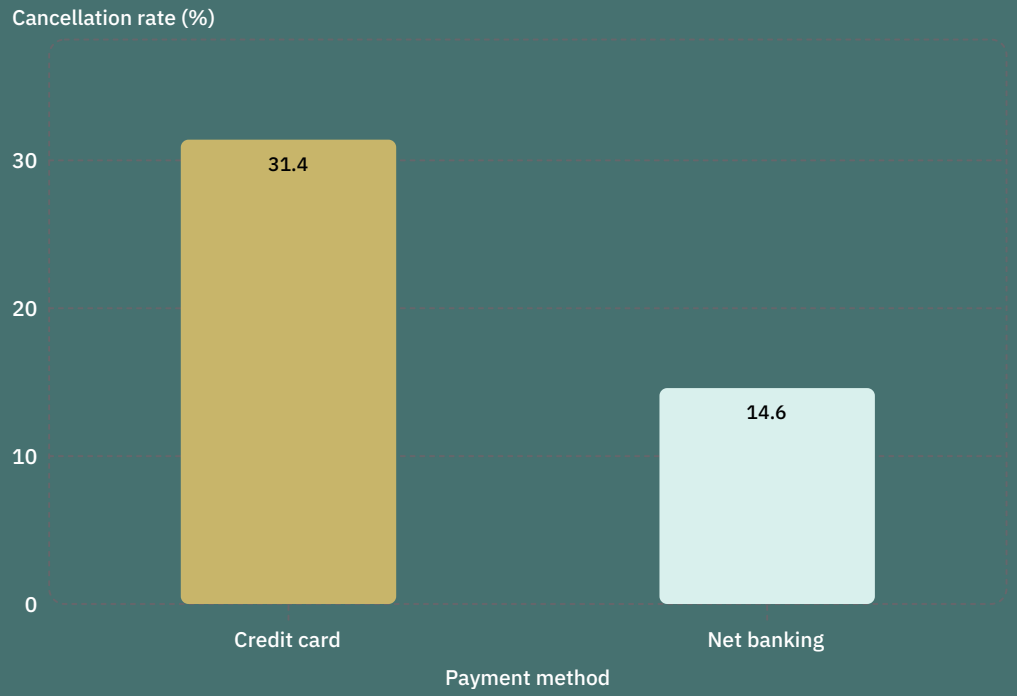
Geographic concentration and ordering rhythm should shape inventory and campaign decisions.

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Returns and cancellations are the clearest leakage points

CANCELLATION RATE
23.0%

RETURN RATE
18.7%



Payment-method comparisons have limited sample sizes and need further investigation.

DELIVERED ORDERS
21.0%

CREDIT CARD
31.4%
cancellation rate

NET BANKING
14.6%
cancellation rate

RATING
3.07
from 45 ratings

37.8% of ratings are 1–2 star; 40% are 4–5 star.

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AVERAGE ORDER VALUE
₹12,768

MEDIAN ORDER VALUE
~₹3,000

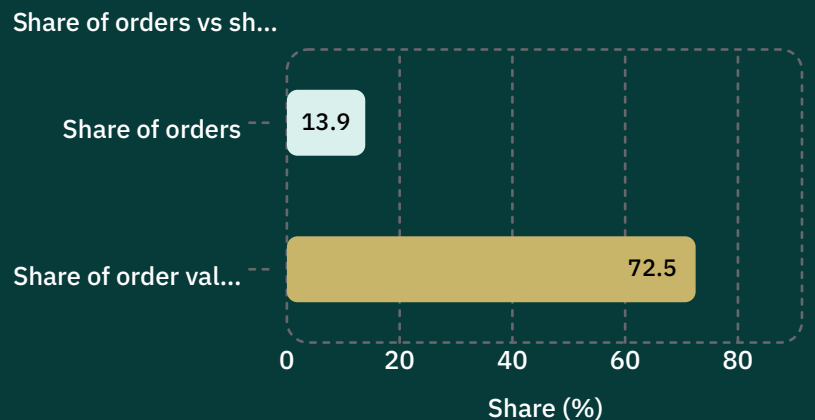
OUTLIERS
35
orders identified

OUTLIER SHARE
13.9%
of orders

VALUE FROM OUTLIERS
72.5%
of total order value

08 / DISTRIBUTION

The average hides a very uneven order book



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09 / DECISION TOOL

Power BI turns analysis into a reusable decision tool

KEY INSIGHTS

Findings and recommendations

SLICERS

year · category · city · payment method

REUSABLE OUTPUT

Analysis becomes a repeatable decision view.

Executive Overview · revenue, orders, AOV, lost percentage, total sale, monthly revenue, status, category, city

Operations & Risk · cancellation, return rate, AOV, outliers, payment method, category, state, weekday, product

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Recommendations: reduce leakage, protect concentration, improve visibility

1. Investigate cancellation and return reasons.
2. Maintain stock availability for high-performing products.
3. Investigate Credit Card payment failures and checkout issues.
4. Explore opportunities to improve sales in Chennai.
5. Test discount strategies before increasing promotional spending.
6. Monitor unusually large orders and report median alongside average order value.

CONCLUSION

Complete analytics workflow from messy data to structured information, meaningful visualization, and actionable business insight.

FUTURE SCOPE

Customer segmentation · sales forecasting · predictive cancellation analysis · larger real-world datasets · real-time dashboard monitoring

THANK YOU

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